RocolSys: An Automatic Row-Column Data Storage System for HTAP

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Abstract. Hybrid Transactional and Analytical Processing (HTAP) brings great challenges to data storage systems. However, traditional approaches have struggled to dynamically adapt storage structures to data and complex workloads. Fortunately, machine learning can provide new methods to guide decisions on data storage structure selection. Motivated by this, we develop RocolSys, an automatic hybrid data storage system for HTAP that can predict workloads and select storage structure automatically based on machine learning. RocolSys predicts workloads accurately and selects storage structures efficiently. It also provides the user-friendly interface that allows users to connect to their own databases. We demonstrate the efficiency of RocolSys on public benchmarks.

Keywords: row and column storage · storage structure selection · workload forecasting · machine learning

1 Introduction

With the development of big data, modern enterprise-class database systems need to support HTAP. HTAP relies on a single system to process the mixed workloads of transactions and analytical queries simultaneously [1]. For the storage structure, row-based storage is suitable for highly concurrent transactional queries, while column-based storage is beneficial for analytic query processing. In order to support both high concurrency and high real-time workloads, the database needs to adopt a row and column coexistence storage structure.

Existing data storage methods support the row storage, column storage or mixed row and column storage [2]. However, the data storage structure mainly dependents on the database administrators (DBAs). Fortunately, machine learning techniques have been used in recent years to address automatic data management techniques [3], which can provide prediction and decision-making services. Thus, we develop RocolSys, an automatic Row-column data storage System For HTAP based on machine learning, which focuses on the following objectives:

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(1) Automatic storage selection. Given the data schema and the workload of the data set, RocolSys can recommend an efficient storage structure for the database based on machine learning, i.e., row-based or column-based storage.

(2) Efficient workload forecasting. RocolSys can predict the workload in the future according to the historical query Logs. It supports dynamic adjustment for the storage structure to adapt to future workload changes.

(3) User personalized selection. RocolSys can show users the execution time of different storage schemes corresponding to the workload. In addition, it provides a variety of time series prediction models. Users can flexibly choose the prediction model and storage structure according to the actual situation.

2 System Overview and Implementation

Figure 1 shows the architecture of RocolSys, including *User Interaction*, *Storage Structure Selection* and *Workload Forecasting*. Considering sufficient storage space, we chose the OpenGauss [4] database as the backend since it has a row storage structure as well as a column storage structure. In addition, users can also replace the backend with their own row-column database.

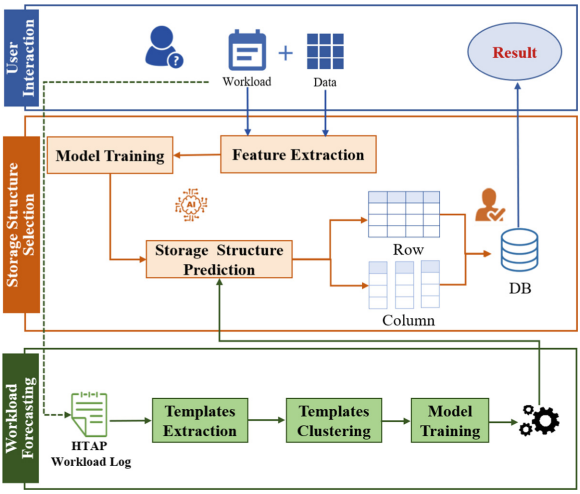


Fig. 1. System Architecture

User Interaction. RocolSys provides friendly interfaces for users to enter data, workloads, and queries. For demonstration purposes, users can compare the performance of machine learning methods with competitors in RocolSys.

Storage Structure Selection. After users upload the data and workload, RocolSys can recommend the storage structure. First, RocolSys can partition

large-scale data and automatically determine the storage structure. To achieve automatic storage, we developed an automatic selection algorithm for storage structure based on machine learning. RocolSys provides a user-friendly interface to determine the storage structure under the guidance of the system.

Workload Forecasting. This component is responsible for predicting future workload based on historical data. It helps users to cope with future workloads changes and adaptively adjust the data storage structure in advance. RocolSys provides a variety of pre-set time series prediction models, which can give the corresponding prediction accuracy results for users to choose.

3 Key Technologies

In this section, we introduce two key machine learning based methods in RocolSys.

Storage Structure Selection. To select the least cost storage structure for the user's data. Firstly, it needs to prepare the data, then divide the data and perform feature selection. RocolSys uses data schema and workload features, including *key field size*, *non-key field size*, *the number of fixed-length fields*, *the number of variable-length fields*, and *the number of rows* involved in a single operation. Then, RocolSys designs the storage decision cost model and uses the performance data to train this model. It analyzes the performance data collected above and designs the cost model. For a given workload and data schema S, the model calculates the cost of row and column storage respectively as shown below:

$$Cost_{row}(S) = W_1 * V_{row-insert}(S) + W_2 * V_{row-select}(S) \tag{1}$$

$$Cost_{column}(S) = W_1 * V_{col-insert}(S) + W_2 * V_{col-select}(S) \tag{2}$$

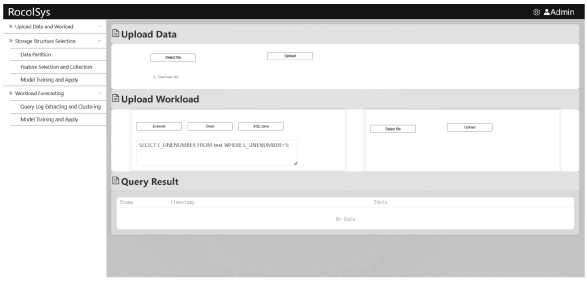
where W_1 denotes the number of queries inserted in the workload, W_2 denotes the number of queries found in the workload, and V_x denotes the predicted value corresponding to the x model ($x =$ insert/select model under row/column store). Finally, RocolSys uses XGBoost [5] to train regression models, which is not only efficient but also lightweight.

Workload Forecasting. To predict the workloads in the future based on historical workloads. The first step is to process the query logs. RocolSys extracts the SQL query from each line of the log file and replaced all numeric constants in the query's SQL string with $. Then RocolSys optimizes DBSCAN to cluster the query templates. RocolSys specifies the distance between object and cluster center, as the evaluation criterion. The DTW distance function is then used as a similarity measure between the historical arrival rates of the two templates. Only when the similarity of query templates within each cluster is high, the prediction arrival rate can be more accurate when the prediction model is deployed for it. RocolSys supports six popular time series prediction algorithms and ensemble learning models to predict query arrival rates [6].

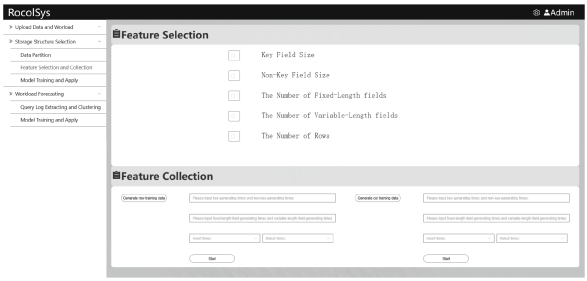
4 Demonstrations

We use the common benchmark set TPC-H [7] to demonstrate the data storage structure recommendation capabilities of RocolSys.

Storage Structure Selection. RocolSys allows users to upload data and workloads as shown in Fig. 2(a). After that, users can choose whether to partition the data or not. Users then choose to use the system in either pre-trained model mode or re-trained mode when using the system. They can interact with the system to perform feature extraction and collection on the data to generate the training set needed for the model as shown in Fig. 2(b). In particular, RocolSys can train the model and display the accuracy of the model to the user, making it easy for the user to adjust and optimize the model parameters based on the model's accuracy. RocolSys can select a storage structure for the given data. When the selection result is displayed to the user, they can decide whether to apply the recommended storage structure or not.



(a) Upload Data and Workload



(b) Feature Selection and Collection

Fig. 2. GUI of RocolSys

Workload Forecasting. RocolSys allows users to customize the selection of query logs for a certain period as historical data for workload prediction, and choose from seven different time series prediction models for training. Similarly, RocolSys can show users the accuracy of different time series prediction models.

It can visually provide the user with prediction results. Based on the workload prediction results, the data that will be accessed frequently in the future can be identified. RocolSys will recommend column stores for this data. RocolSys can get around 85% performance improvement for users by the model of storage structure recommendation.

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